

2.3 Second-Order Methods

Newton, Levenberg–Marquardt, and BFGS.

1. Newton

Approximate J by a quadratic at the current point $\theta^{(k)}$:

$$J(\theta^{(k)} + \delta) \approx J(\theta^{(k)}) + \delta^T \nabla J(\theta^{(k)}) + \frac{1}{2} \delta^T H(\theta^{(k)}) \delta,$$

where $H = \nabla^2 J$. Minimize that quadratic: solve

$$H(\theta^{(k)}) \delta = -\nabla J(\theta^{(k)}),$$

then $\theta^{(k+1)} = \theta^{(k)} + \delta$. On a convex quadratic, **one** Newton step is exact. That is the point of Lab 2, exercise 3: a badly scaled bowl that GD crawls and Newton finishes.

2. When Newton breaks

If H is not positive definite, the quadratic is not a bowl and the step can go uphill or blow up. A cheap fix is to solve $(H + \lambda I) \delta = -\nabla J$ (**Levenberg–Marquardt**). Large λ looks like gradient descent. Small λ looks like Newton.

Analytical second derivatives are often unpleasant. Finite differences are possible and do not guarantee an SPD H .

3. Quasi-Newton and BFGS

Do not form H . Keep a symmetric positive-definite **stand-in** H_* for H^{-1} , update it from the observed $\Delta \theta$ and $\Delta(\nabla J)$, and set

$$\delta^{(k)} = -H_* \nabla J(\theta^{(k)}),$$

then line-search. **BFGS** is the update you should be able to name. Rank-two, stays SPD, no explicit Hessian. Faster and more stable than raw Newton on many medium-scale problems; still heavier than SGD on a million-example net.

Module 6 will come back to first-order methods (Adam, RMSProp). The idea is the same: better than raw $-\nabla J$, cheaper than H^{-1} .